

POLYMARKET BTC BOT v6.2

Complete Guide — How It Works



Every 2 seconds — 24/7 automated trading loop

Version	v6.2 — Intelligent Trading Engine
Market	Polymarket BTC 15-Minute Up/Down
Strategies	5 (ARB, Latency, Momentum, Flash, Squeeze)
Sizing	Dynamic — % of balance, auto-scales
Risk Layers	5 protection layers

1. What The Bot Does

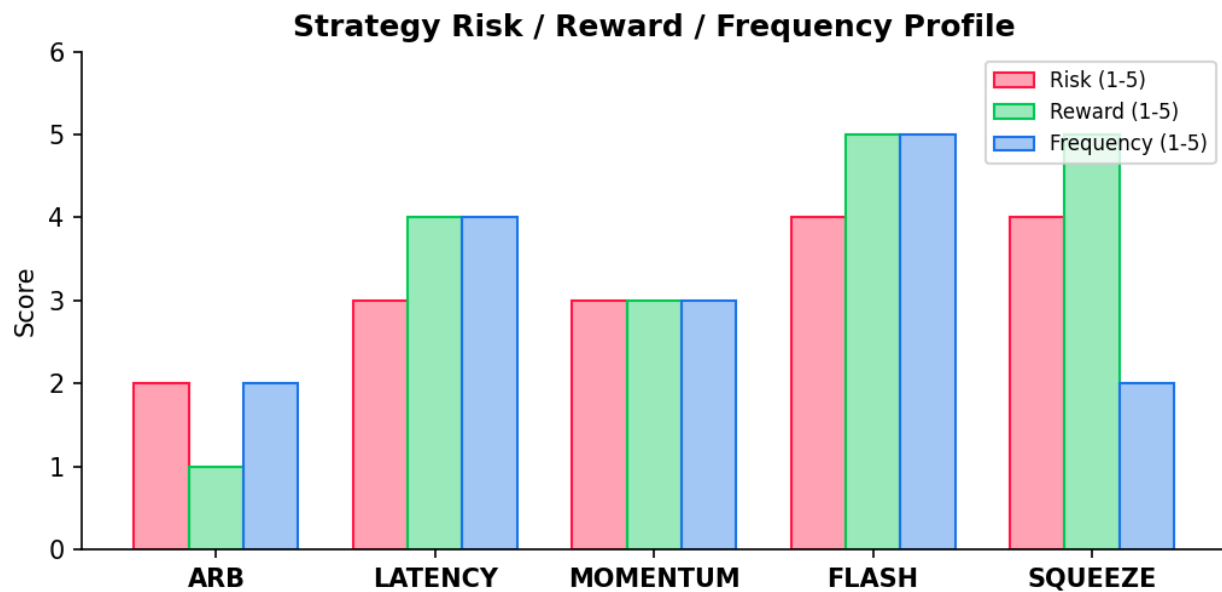
Polymarket runs 15-minute binary markets on Bitcoin. Every 15 minutes a new market opens asking: **"Will BTC go UP or DOWN in the next 15 minutes?"** You can buy YES tokens (betting UP) or NO tokens (betting DOWN). If you're right, each token pays \$1.00. If wrong, you lose your bet.

The bot watches Bitcoin's price on Binance every 2 seconds and compares it to what Polymarket thinks. When it spots a mismatch — like BTC clearly rising but YES tokens still cheap — it buys automatically. It runs 24/7 and trades every market cycle.

The Trading Loop

Step	What Happens	Frequency
1. Price Feed	Gets BTC price from Binance (fallback: Coinbase)	Every 2 sec
2. Market Scan	Finds current 15-min BTC market on Polymarket	Every 10 sec
3. Trend Update	Analyzes BTC direction, momentum, regime	Every 5 sec
4. Strategy Check	Runs all 5 strategies looking for signals	Every 10 sec
5. Risk Check	Verifies balance, position limits, loss limits	Before each trade
6. Execute	Places order on Polymarket via CLOB API	When signal found
7. Monitor	Tracks open positions, resolves when market ends	Continuous
8. Auto-Redeem	Claims winnings back to USDC balance	Every 5 min

2. The 5 Trading Strategies



Strategy 1: ARB (Arbitrage)

Buys whichever side is cheaper when YES + NO prices add up to less than \$0.99. In a perfect market, YES + NO = \$1.00 (minus fees). When the sum drops below \$0.99, there's free money — buy the cheap side and you're almost guaranteed profit when the market resolves. This is the lowest-risk strategy.

Trigger	YES + NO < \$0.99
Entry Price	\$0.10 - \$0.55
Risk Level	Low — mathematical edge
Sizing	5% of balance

Strategy 2: LATENCY (Speed Advantage)

Exploits the speed gap between Binance (updates instantly) and Polymarket (updates slowly). When BTC jumps 0.3%+ on Binance but Polymarket prices haven't moved yet, the bot buys the correct side before other traders react. This is the highest-conviction strategy because you're trading on information that hasn't been priced in yet.

Trigger	BTC moves 0.3%+ but Polymarket lags
Min Edge	8% (predicted probability - market price)
Trend Boost	+3% edge when trend agrees, -3% when against
Risk Level	Medium — strong when trend confirms
Sizing	6% of balance (highest allocation)

Strategy 3: MOMENTUM (Technical Analysis)

Uses 5 technical indicators to predict BTC's short-term direction: Bollinger Bands, EMA crossover, RSI, Rate of Change, and VWAP. When all indicators align (65%+ confidence) and VWAP confirms, the bot bets on the predicted direction. Requires trend agreement.

Indicators	BB (30%), EMA (20%), RSI (20%), ROC (15%), VWAP (15%)
Min Confidence	65% composite score
VWAP Check	Must agree with overall direction
Risk Level	Medium — multiple confirmations required
Sizing	5% of balance

Strategy 4: FLASH (Buy Cheap Tokens)

Buys tokens when they drop to very low prices (\$0.05-\$0.27). The idea: markets sometimes overshoot — a side drops to \$0.10 when BTC barely moved. If you buy at \$0.10, you only need it to win 10% of the time to break even (\$1.00 payout). In practice, these cheap tokens win 25-35% of the time, making this highly profitable despite the low win rate.

Trigger	YES or NO token drops below \$0.27
Confirmation	BTC momentum in your direction + trend support
Best Entries	\$0.05-\$0.15 (5-20x payout)
Risk Level	Higher per trade — but asymmetric reward
Sizing	4% of balance

Strategy 5: SQUEEZE (Volatility Breakout)

Detects when BTC volatility contracts (Bollinger Bands squeeze) then explosively expands. This pattern often precedes big moves. When the squeeze fires and BTC picks a direction, the bot buys that side at a cheap price before the market fully reacts.

Trigger	Bollinger Band width contracts then expands
Confirmation	BTC breaks out in clear direction
Risk Level	Higher — but catches big moves
Sizing	5% of balance

3. The Trend Engine (AI Brain)

The Trend Engine is the shared intelligence layer that all strategies consult before trading. It analyzes BTC's behavior across multiple timeframes and categorizes the current market into a "regime" that determines which strategies are allowed to fire and how aggressively.

Trend Directions

Direction	Condition	Effect
UP	BTC 2-min change > +0.08%	Boost YES trades, block NO
DOWN	BTC 2-min change < -0.08%	Boost NO trades, block YES
DRIFT_UP	BTC 30-sec change > +0.03%	Mild YES preference
DRIFT_DN	BTC 30-sec change < -0.03%	Mild NO preference
FLAT	Neither condition met	Only ARB trades allowed

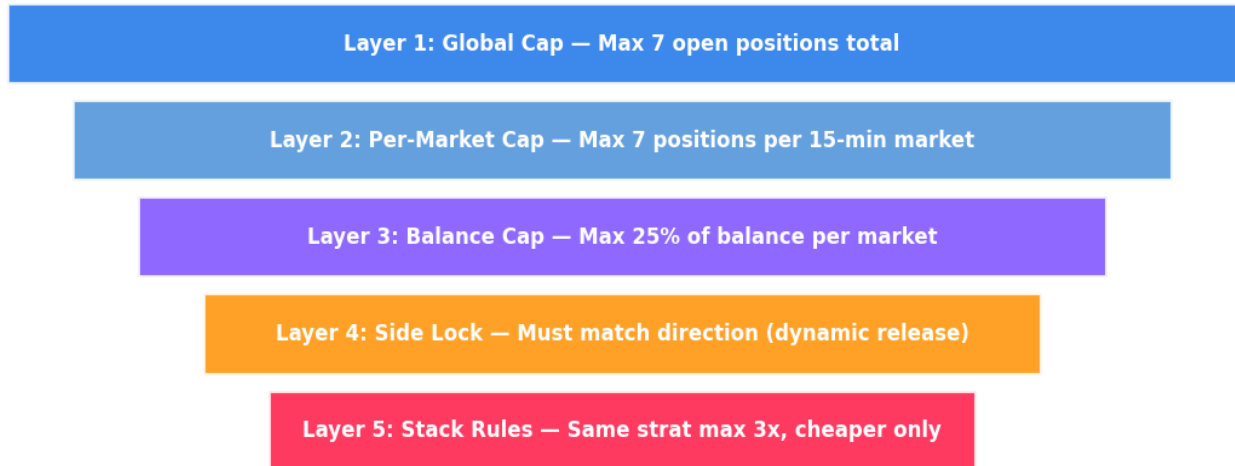
Market Regimes

Beyond simple direction, the engine classifies the market into regimes that control position stacking:

Regime	What It Means	Stacking Allowed?
RANGING	BTC bouncing in a range	No — first entry only
TRENDING_UP	Clear sustained uptrend	Yes — can add more YES
TRENDING_DOWN	Clear sustained downtrend	Yes — can add more NO
BREAKOUT	Explosive move starting	Yes — catch the wave
CHOPPY	Random noise, no direction	No — minimal trading

4. Risk Management — 5 Protection Layers

Risk Protection — 5 Layers Deep



Layer 1: Global Position Cap

Maximum 7 open positions at any time across all markets. Once you hit 7, no new trades until one resolves.

Layer 2: Per-Market Cap

Maximum 7 positions on a single 15-minute market. Prevents over-concentrating on one outcome.

Layer 3: Balance Cap (25%)

No more than 25% of your total balance can be at risk on a single market. At \$400 balance, that's \$100 max per market. This ensures one bad market can't wipe you out.

Layer 4: Dynamic Side Lock

The first trade in a market sets the direction (YES or NO). All subsequent trades must match. This prevents the bot from betting both sides and guaranteeing a loss. The lock can release if the trend clearly reverses, existing risk is small (<8%), and there's 7+ minutes left.

Layer 5: Strategy Stacking Rules

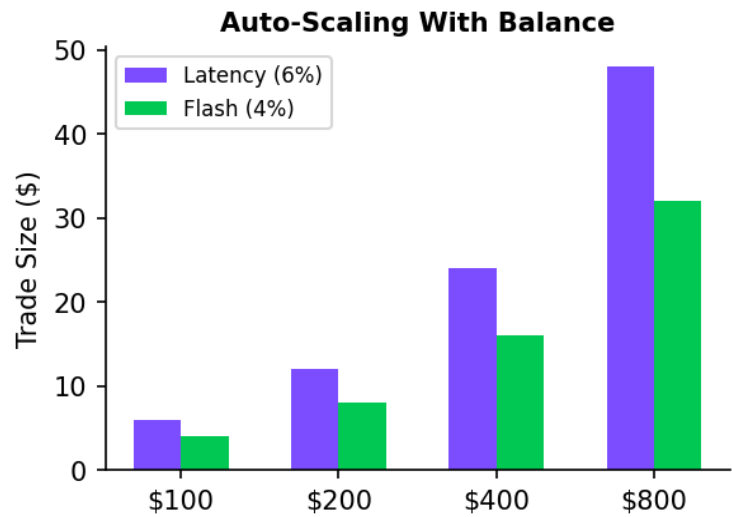
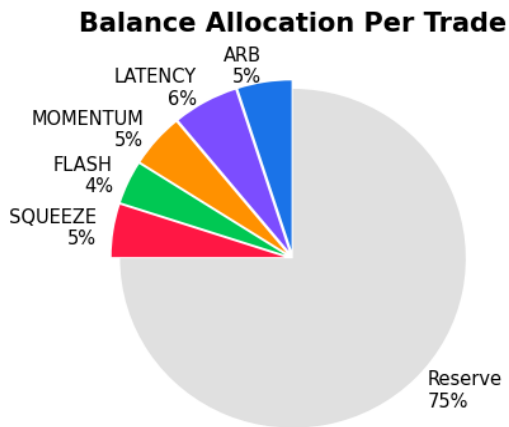
Same strategy can stack up to 3 times per market, but ONLY when the market is trending/breaking out, 5+ minutes remain, and the new entry price is cheaper than the existing one (averaging down, not up). Different strategies joining on the same side is always allowed — that's a strong confirmation signal.

Additional Safety Features

Feature	How It Works
Daily Loss Limit	Bot stops trading if losses exceed \$50 (configurable)

Bad Hour Detection	Tracks win rate per hour; pauses if < 25% over 3+ trades
Losing Streak Pause	4+ consecutive losses = all strategies pause
Spread Check	Won't trade if bid-ask spread > 8% (bad liquidity)
Min Time Left	Won't enter with < 90 seconds until market expires
Auto-Redeem	Claims winnings every 5 minutes back to USDC

5. Dynamic Position Sizing



The bot sizes every trade as a percentage of your current balance. As your balance grows, trade sizes grow automatically. As it shrinks, sizes shrink to protect you.

Strategy	% of Balance	At \$144	At \$400	At \$1000
ARB	5%	\$7.20	\$20.00	\$50.00
LATENCY	6%	\$8.64	\$24.00	\$60.00
MOMENTUM	5%	\$7.20	\$20.00	\$50.00
FLASH	4%	\$5.76	\$16.00	\$40.00
SQUEEZE	5%	\$7.20	\$20.00	\$50.00
TOTAL (all 5)	25%	\$36.00	\$100.00	\$250.00

Adaptive Size Multipliers

The bot also tracks each strategy's win rate and adjusts sizing with multipliers. A strategy winning 60%+ gets up to 2.5x its base size. A strategy losing gets reduced to 0.3x. This means the bot automatically puts more money behind what's working and less behind what isn't.

6. How Profits Work

The Math Behind Each Trade

Every trade is a binary bet. You buy tokens at a price between \$0.01 and \$0.99. If your side wins, each token pays \$1.00 (minus 2% fee). If it loses, you lose your bet.

	Example 1: Win	Example 2: Loss
Buy	25 NO tokens @ \$0.20 = \$5.00	25 YES tokens @ \$0.30 = \$7.50
Outcome	BTC goes DOWN — NO wins!	BTC goes DOWN — YES loses
Payout	$25 \times \$1.00 \times 0.98 = \24.50	\$0.00
Profit	$\$24.50 - \$5.00 = +\$19.50$	-\$7.50
Return	+390%	-100%

Why Low Win Rate Still Profits

The bot's real-world performance showed a 29% win rate but was still profitable (+\$24 overnight). This works because of **asymmetric payoffs**: average win (\$33) is 3x the average loss (\$12). At these ratios, you only need a 23% win rate to break even.

Avg Entry Price	Payout If Win	Breakeven Win Rate
\$0.10	\$0.98 (9.8x)	10%
\$0.15	\$0.98 (6.5x)	15%
\$0.20	\$0.98 (4.9x)	20%
\$0.25	\$0.98 (3.9x)	26%
\$0.30	\$0.98 (3.3x)	31%
\$0.40	\$0.98 (2.5x)	41%
\$0.50	\$0.98 (2.0x)	51%

Green rows = sweet spot. Bot targets entries at \$0.10-\$0.25 where the math is strongest.

7. Setup & Running

Files Needed

File	Purpose
polymarket_btc_bot.py	The bot — single file, everything included
.env	Your private key, wallet address, settings

Auto-Generated Files

File	Purpose
trade_history.txt	Human-readable log of all trades
trade_data.json	Machine-readable trade data
traded_cids.json	Tracks which markets need redemption
polybot.log	Debug log for troubleshooting

Commands

Action	Command
Install dependencies	<code>pip install py-clob-client python-dotenv requests numpy colorama web3</code>
Run the bot	<code>python3 polymarket_btc_bot.py</code>
Run in background (Linux)	<code>screen -S bot python3 polymarket_btc_bot.py (Ctrl+A then D to detach)</code>
Reattach to see it	<code>screen -r bot</code>
Stop the bot	<code>Ctrl+C</code> (shows session summary)

Dashboard Preview

The bot displays a live terminal dashboard showing: connection status, balance, P&L, trend direction, streak count, side lock status, all 5 strategy statuses with dynamic bet sizes, open positions with progress bars, and trade history (newest first, last 10 trades).

Bot auto-detects balance from Polymarket on startup. No manual configuration needed — just set your private key and wallet address in .env and run.